(25) Myhill Nerode Theorem - Table filling Method

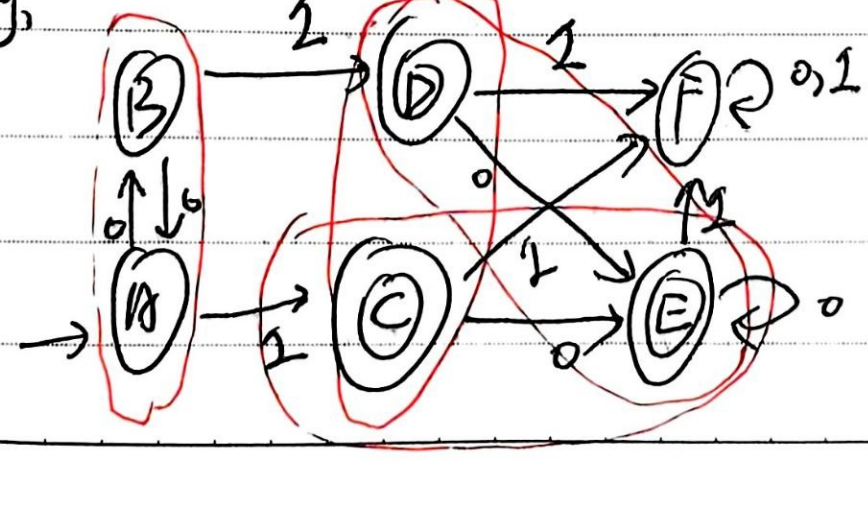
5.3

for minimization of DFA

Steps:

1) Draw a table for all pairs of states (P, Q)
2) Mark all pairs where $P \in F$ and $Q \notin F$
3) if there are Unmarked pairs (P, Q) such that $[\delta(P, X), \delta(Q, X)]$ is marked, then mark [P, Q], REPEAT THIS UNTIL NO MORE MARKINGS CAN BE MADE
4) Combine all the Unmarked Pairs and make them a single state in the minimized DFA

eg,



Mo | Tu | We | Th | Fr | Sa | Su

Memo No. ________

Date / /

⇒

	A	B	C	D	E	F
A						
B	0					
C	✓	✓				
D	✓	✓	0			
E	✓	✓	0	0		
F	✓	✓	✓	✓	✓	

Step 3: (B, A) − $\delta(B, 0) : A$, $\delta(A, 0) : B$ } not need to mark; $\delta(B, 1) : D$, $\delta(A, 1) : C$ } not need to mark

✓

→ (D, C) − $\delta(D, 0) : E$, $\cancel{\delta(D)}$ $\delta(C, 0) : E$ }; $\delta(D, 1) = F$, $\delta(C, 1) = F$ } ✓

→ (E, C) − $\delta(E, 0) : E$, $\delta(C, 0) : E$; $\delta(E, 1) : F$, $\delta(C, 1) : F$ ✓

→ (F, A) − $\delta(F, 0) : F$, $\delta(A, 0) : B$; $\delta(F, 1) : F$, $\delta(B, 1) : D$ } X ⇒ need to mark

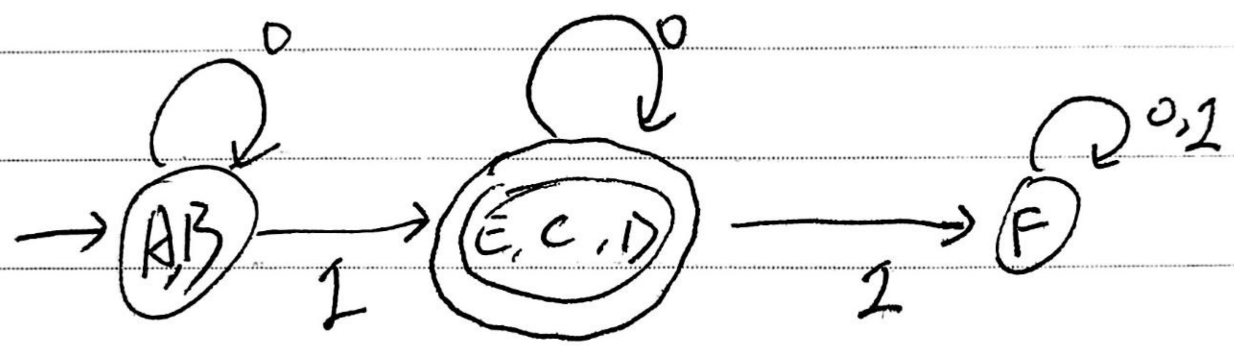
→ (E, D) − $\delta(E, 0) : E$, $\delta(D, 0) : E$; $\delta(E, 1) = F$, $\delta(D, 1) = F$

→ (F, B) − $\delta(F, 0) = F$, $\delta(B, 0) = A$ ⇒

Step 4:

$\underset{}{(A,B)},\ \underbrace{(C,D),\ (E,C),\ (E,D)}_{\to (E,C,D)}$

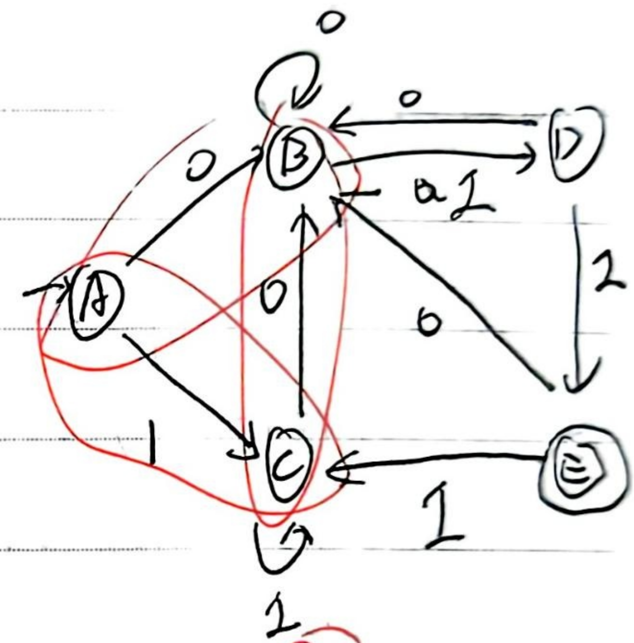
$\Rightarrow$ combine to one step



$\boxed{\text{Myhill Nerode Theorem}}$

26. Example of Myhill Nerode Theorem

	A	B	C	D	E
A					
B	✓				
C		✓			
D	✓	✓	✓		
E	✓	✓	✓	✓	



Step 3: (B, A). $\delta(B,0): B$, $\delta(A,0): B$, $\delta(B,1): D$, $\delta(A,1): C$ ✓

(C, A): $\delta(C,0): B$, $\delta(A,0): B$, $\delta(C,1): C$, $\delta(A,1): C$ ✓

(C, B): $\delta(C,0): B$, $\delta(B,0): B$, $\delta(C,1): C$, $\delta(B,1): D$ ✓

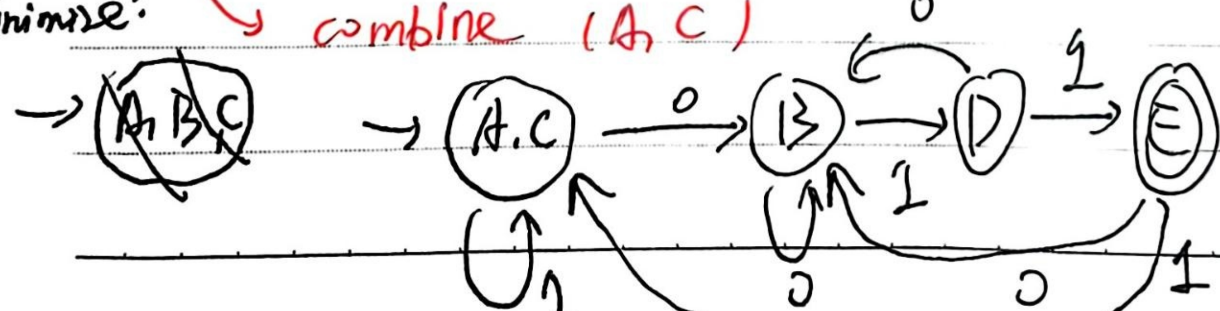
(D, A): $\delta(D,0): B$, $\delta(A,0): B$, $\delta(D,1): E$, $\delta(A,1): C$ ✓

(D, B), $\delta(D,0): B$, $\delta(B,0): B$, $\delta(D,1): E$, $\delta(B,1): D$

(D, C): $\begin{cases} B \\ B \end{cases}$, $\begin{cases} E \\ C \end{cases}$

Step 4: ~~combin (A,B) (A,C), (B,C) ⇒ (A,B,C)~~

minimize: combine (A, C)



Mo | Tu | We | Th | Fr | Sa | Su

Memo No. ______

Date / /

(27) Finite Automata with Outputs

MEALY MACHINE

MOORE MACHINE

$(Q, \Sigma, \Delta, \delta, \lambda, q_0)$

where

Q = Finite Set of Sates

Σ = Finite non-empty set of Input Alphabets

Δ = The set of Output Alphabets

δ = Transition function : $Q \times \Sigma \to Q$

λ = Output function : $\boxed{\Sigma \times Q \to \Delta}$

q_0 = Initial State / Start State